

Coding Challenge

Overview

Imagine that we are building a REST API for a conversational AI virtual agent that can answer questions about movies using an open movie dataset. The agent should leverage both structured data retrieval and LLM-powered response generation to provide informative and conversational answers. Your task in this coding challenge is building an application that solves this problem.

Requirements

Data Setup

- Use the **MovieLens dataset** (small version: ~100k ratings, ~9k movies) or **TMDB 5000 Movie Dataset**
- Store movie data in a local SQLite database with tables for movies, genres, and ratings
- Include fields: title, year, genres, overview/plot, ratings, cast, director

Core Functionality

1. **Data Ingestion:** Create a script to load the dataset into your local database
2. **Query Processing:** Implement intelligent query parsing to identify movie-related intents (e.g., "recommend action movies from 2020", "tell me about Inception")
3. **LLM Integration:** Use an open-source LLM (via Hugging Face Transformers, Ollama, or OpenAI API) to generate natural, conversational responses based on retrieved data

Technical Requirements

- Python 3.11+ with FastAPI or Flask
- SQLite for local data storage
- LLM Integration using transformers, openai, or similar library
- Basic logging, error handling, and unit/integration testing as you find necessary. We don't expect complete test coverage

Evaluation Criteria

- Code quality and structure
- API design and documentation (define the endpoints according to the requirements)
- Effective combination of structured data retrieval with LLM responses
- Clear setup instructions

Deliverables

- We do not expect you to spend more than a few hours on this challenge. Consider this a proof of concept. Not everything has to be perfect. If you don't have time to finish everything, please explain which parts are incomplete
- Naturally, StackOverflow and coding assistants like ChatGPT exist. While we cannot prevent you from using these, we expect that you fully own any code you submit and can talk to it comprehensively in a technical interview. Knowing your dependencies and the rationale of design decisions is crucial in your day-to-day work, and we expect the same of you in your coding challenge
- Source code with *requirements.txt* in a public repository (e.g., GitHub, GitLab). (Please make the repository name a random string and avoid references to Genesys to prevent others copying your code in the future)
- Include some basic setup and usage instructions on how to run the application. Sample API requests/responses are also helpful for the reviewers
- Brief explanation of your approach to combining structured data with LLM capabilities
- If you have questions or clarifications, please contact your recruiter
- During the technical interview, we will be discussing the code you submitted, so be prepared to talk in-depth about the implementation, enhancements, etc.
- Have fun!